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# Rakshasutra: a privacy-preserving, edge-filtered BLE and LoRaWAN architecture for finding separated pilgrims on mountain routes without cellular coverage

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**Abstract.** Pilgrimage routes in the mountains are hard places to find a lost child or a collapsed elder. Cellular networks congest or disappear behind ridges, long steel rain shelters block radio, and phones must be left in cloakrooms before the inner shrine. Passive RFID cards see a pilgrim only within 1 to 3 m of a gate, and crowd-sourced Bluetooth trackers need other people's phones and a data connection. This paper describes Rakshasutra, a design that needs neither: a low-cost Bluetooth Low Energy wristband that broadcasts a signed, rotating pseudonym, and pole-mounted ESP32-S3 and SX1262 relays that filter what they hear and forward it over LoRaWAN in the Indian 865–867 MHz band, across a 22.5 km route modelled on the Shri Mata Vaishno Devi trek with 255 poles. The wristband sends a 15-byte advertisement whose 32-bit identifier and radio address change every 15 minutes and whose 32-bit AES-CMAC lets the server reject forged alarms. Each pole reports an arrival once, in chunks of at most 25 wearables, and summarises who is still present as a fixed 64-byte Bloom filter, so no frame exceeds the 222-byte limit however dense the crowd. **Nothing in this paper has been built or measured.** Every result is calculated from the stated model or simulated, by open code that reproduces each number. For a surge of 3,000 wearables in one gateway cell the model gives 30.5% uplink frame loss for the design as specified and 21.3% once departure lists are dropped, against 57.6% for one uplink per arrival; a simulated SOS reaches the control room in a median of 0.5 s, and within 4 s in 92% of cases. The audit behind this revision also found three mechanisms of the first draft that do not work as specified, and states the fix for each.

**Keywords:** AES-CMAC, BLE Coded PHY, Bloom filter, crowd localisation, edge filtering, LoRaWAN IN865, privacy, pseudonymous authentication.

## 1 Introduction

The Shri Mata Vaishno Devi shrine in the Trikuta hills of Jammu and Kashmir receives about 9.5 to 10 million pilgrims a year, who walk or ride some 22.5 km of steep, branching mountain path. Children under twelve and people over sixty are the most likely to be separated from their group or to need help between checkpoints. Four conditions on such a route defeat the usual tools.

1. **Consumer Bluetooth trackers need strangers' phones.** Tags such as the Apple AirTag carry no satellite receiver or modem [12]. They are located when a nearby phone hears them and reports over the internet. On the trek the data network is congested or shadowed, and there is no common app.
2. **Passive RFID sees only the gates.** The *Yatra Parchi* card issued at registration is powered by the reader and answers within 1 to 3 m of a turnstile. Between gates it says nothing.
3. **Steel shelters block radio.** Much of the path runs under galvanised iron (GI) sheds that protect pilgrims from rain and rockfall. A grounded corrugated roof attenuates signals that try to pass through it.
4. **Phones are not allowed near the sanctum.** Phones are deposited in cloakrooms before the Bhawan complex. A guard who finds a lost child cannot call a parent whose phone is in a locker.

Rakshasutra, named after the protective thread tied on the wrist, turns the problem round. Instead of a tracker that finds its own position, a reusable wristband costing an estimated INR 225–235 only announces itself, and poles already standing along the shelters, 25 to about 110 m apart, hear it and report to an on-premise server.

### Contributions

- **A signed, pseudonymous advertisement.** A 15-byte frame carrying a 16-bit slot counter, a 32-bit identifier that rotates every 15 minutes, and a 32-bit truncated AES-CMAC [3]. The server resolves it with one hash lookup; poles hold no keys.
- **An edge filter with bounded airtime.** A pole reports each wearable once on arrival, in chunks of at most 200 bytes, and replaces the list of those present with a 64-byte Bloom filter [5, 6].
- **Alerts that keep up with people.** A direct tether between a child's band and a parent's tag, and a Class C multicast of a wake token to the poles around a wearable, in place of slow Class A downlinks.
- **Capture of passengers in electric vehicles**, by a fixed 800 ms advertising interval and a single-frame arrival rule on transit tracks, and a split-antenna pole that keeps both radios clear of the roof.
- **An executable audit.** Code that recomputes every number here, and a list of what the first draft got wrong (Section 9).

> **Status of the evidence.** This is a design. The propagation figures (roof loss, path-loss exponent, body loss) are design values that have not been measured on the trek; currents are from the first draft's power table. Results are labelled *calculated* when they follow from a formula and *simulated* when they come from a Monte Carlo run. Code, parameters and results: https://github.com/0xr4j/rakshasutra.

## 2 Related work

**Printed QR bands and passive RFID** cost under INR 10 and are purely reactive: a child is found only at a reader or when someone scans the band. **Cellular and satellite trackers** give continuous positions under open sky, but suffer sky blockage and multipath under steel shelters, depend on base stations that saturate in a surge, cost over INR 2,500 with a SIM, and last a day or two. **Crowd-sourced tags** are cryptographically careful [12] but need dense, connected smartphones. **Camera-based recognition** struggles with shawls, caps and dim shelters. **Earlier BLE-to-LoRaWAN proposals** forward raw beacon addresses and signal strengths through Class A nodes [9, 11]; in a dense crowd this exceeds payload limits, collides heavily, exposes a fixed address to anyone listening, and cannot deliver a timely downlink. Table 1 summarises.

## 3 System model

### 3.1 The route as a graph

The route is a weighted graph $G = (V, E, W)$. $V$ is the set of $N = 255$ poles, at altitudes between 762 m and 2,017 m. $E$ joins adjacent poles along six zones: zone 1, the old track and its stairways (6.0 km); zone 2, Tarakote Marg (7.0 km); zone 3, the Adhkuwari junction (0.5 km); zone 4, the Himkoti electric-vehicle track and the Sanjichhat track (6.5 km together); zone 5, the Bhawan complex, skywalk and cloakrooms (1.0 km); and zone 6, the climb to Bhairon Ghati (1.5 km). $W$ gives each edge its length in metres. The route splits at Banganga and again at Adhkuwari, so the difference of two pole numbers says little about how far apart two people are. Separation is the shortest path:

$$d_G(p_a, p_b) = \min_{\pi \in \Pi(p_a, p_b)} \sum_{e \in \pi} w(e). \tag{1}$$

**Table 1** How the approaches compare on a covered mountain route. Entries for Rakshasutra are design targets and calculated values, not measurements.

| | Passive RFID or QR | GPS and cellular tracker | Consumer BLE tag | BLE with raw LoRaWAN forwarding | Rakshasutra |
|---|---|---|---|---|---|
| Tracking between gates | No | Open sky only | Needs nearby phones | At low density | Yes, to the nearest pole |
| Works without cellular | At gates | No | No | Yes | Yes |
| Works under steel sheds | At gates | No | Poor | Poor with one antenna | Yes, with split antennas |
| Works where phones are banned | No alert path | – | No | No alert path | Kiosks and the parent tag |
| Resists stalking and replay | Static ID | SIM encryption | Rotating key | None | Rotating pseudonym, signed SOS |
| Loss with 3,000 tags in a cell | – | – | – | 57.6% (calculated) | 21.3–30.5% (calculated) |
| Wearable battery | None | 1–2 days | About a year | Months | 9.0–12.8 months on Coded PHY (calculated) |

In the reference layout the last pole of the Himkoti track and the first pole of the Sanjichhat track carry consecutive numbers and are 3.0 km apart by track. The 60 poles of zones 3 and 5 use the shrine's fibre network; the other 195 use LoRaWAN, which over 21.0 km of track is one pole per 108 m on average.

### 3.2 Radio under a steel roof

Pilgrims walk under GI shelters about 3.2 m high and 4.5 m wide. Let $L_{GI}(f)$ be the loss through the roof. The design values are 18.4 dB at 866 MHz and 24.1 dB at 2.44 GHz; both await measurement. A pole mounted wholly above the roof loses 24.1 dB on the Bluetooth link from a wrist below, on top of body shadowing of 8 to 16 dB. A pole wholly below the roof loses 18.4 dB on its LoRa link to a hilltop gateway. Rakshasutra splits the two.

**Bluetooth below the roof.** The ESP32-S3 enclosure is clamped under the roof truss at 3.0 m with a clear view down. The received power at distance $d$ follows a log-distance model:

$$P_{rx}(d) = P_{tx} + G_{tx} + G_{rx} - PL(d_0) - 10\,n \log_{10}(d/d_0) - L_{body} - X_{\sigma}, \tag{2}$$

with $P_{tx} = +8$ dBm, $PL(d_0{=}1\,\mathrm{m}) = 40.2$ dB, exponent $n = 2.15$ under the roof, $L_{body} = 14$ dB in a heavy crowd, and shadowing $X_{\sigma} \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 4.2$ dB. Antenna gains are taken as 0 dB. With Coded PHY ($S = 8$, sensitivity $-103$ dBm) [4] the margin at 50 m is 20.3 dB, and 98% of frames clear the sensitivity out to 174 m (82 m on 1M PHY). A pole above the roof with a $+4$ dBm, 1M PHY wearable reaches 4.1 m.

**LoRa above the roof.** One metre of RG-316 (0.9 dB) carries the SX1262 output through a gland to a 3 dBi whip above the ridge. With $P_{tx} = +14$ dBm, a gateway antenna of 5.8 dBi and an SX1302 sensitivity of $-126.5$ dBm at SF7, the maximum allowable path loss is 145.4 dB. The paper does not fix a path-loss model for the valley, so no range is claimed. As an indication, free-space loss plus one ridge of an assumed 20 dB leaves a margin of 16.1 dB at 8 km with the whip above the roof, and $-4.4$ dB with an internal antenna below it.

## 4 Pseudonymous protocol

**Table 2** Link budget recomputed from Eq. 2 and the stated parameters (calculated, not measured). Bluetooth rows give the distance at which 98% of frames clear the receiver sensitivity; LoRa rows give the margin at 8 km behind one assumed 20 dB ridge.

| Link | PHY | $P_{tx}$ | Roof | $S_{rx}$ | Range or margin |
|---|---|---|---|---|---|
| Wearable to pole, pole above the roof | BLE 1M | +4 | 24.1 | -96.0 | 4.1 m |
| Wearable to pole, pole under the roof | BLE 1M | +8 | 0.0 | -96.0 | 82.2 m |
| Wearable to pole, pole under the roof | BLE Coded S=8 | +8 | 0.0 | -103.0 | 174.1 m |
| Pole to gateway, antenna under the roof | LoRa SF7 | +14 | 18.4 | -126.5 | -4.4 dB |
| Pole to gateway, whip above the roof | LoRa SF7 | +14 | 0.0 | -126.5 | +16.1 dB |

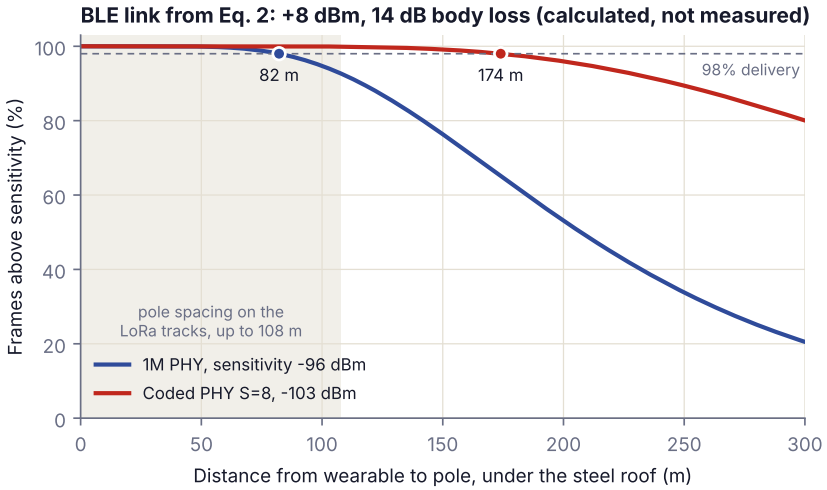



**Figure 1** Share of frames above sensitivity against distance, from Eq. 2.

### 4.1 Threats

Four adversaries are considered. $\mathcal{A}_1$ listens along the route and tries to follow one child or to link a family. $\mathcal{A}_2$ records advertisements and plays them back elsewhere or later. $\mathcal{A}_3$ sends forged commands to make a band buzz or to drain it. $\mathcal{A}_4$ steals a pole or a band and reads its memory.

### 4.2 Keys

The server keeps a 128-bit master key $K_{\text{master}}$ in an encrypted keystore or a hardware security module. Each wearable $u$ is given, at manufacture, its own key

$$K_u = \text{CMAC}_{K_{\text{master}}}(\text{UID}_u \,\|\, \text{Salt}) \,, \tag{3}$$

and its debug port is locked. Poles never hold $K_{\text{master}}$ or any $K_u$. They authenticate to the network server with their own LoRaWAN keys, protected by secure boot and flash encryption, and a tamper switch clears their memory.

### 4.3 The 15-byte advertisement

A wearable keeps a 16-bit counter $c$ driven by its 32.768 kHz crystal and increases it every 900 s. In each slot it derives a fresh identifier and a fresh radio address:

$$\mathrm{ID}_{u,c} = \mathrm{Trunc}_{32}\,\mathrm{CMAC}_{K_u}(\texttt{0x01} \,\|\, c)\,, \tag{4}$$

$$M_{u,c} = \mathrm{Trunc}_{46}\,\mathrm{CMAC}_{K_u}(\texttt{0x02} \,\|\, c) \text{ with both top bits set.} \tag{5}$$

It broadcasts the payload of Table 3 every 800 ms, and every 200 ms while SOS is pressed. Because the counter is sent, nothing needs to be written to the wearable at registration. Three encodings left open in the first draft are fixed by the reference code: truncation keeps the leading bytes of the CMAC, the counter enters each CMAC as the same little-endian bytes it has on air, and the company identifier is sent low byte first.

**Table 3** The signed advertisement payload.

| Bytes | Field | Content |
|---|---|---|
| 0–1 | Company_ID | `0xFF5A`, a laboratory value |
| 2 | V_type | bits 0–3 version (4); bit 4 role: band or parent tag |
| 3–4 | Slot_Counter | $c$, little-endian |
| 5–8 | Rolling_ID | $\mathrm{ID}_{u,c}$ |
| 9 | F_stat | bit 0 SOS; 1 tether lost; 2 buzzer on; 3 low battery |
| 10 | B_mv | supply voltage in 20 mV steps |
| 11–14 | MIC | $\mathrm{Trunc}_{32}\,\mathrm{CMAC}_{K_u}$(bytes $2 \ldots 10$) |

### 4.4 Resolving an identifier

At the registration counter the operator scans the family's card and the codes on the band and the parent tag, and rests both on a listener pad. Lifting a band off its storage magnet wakes it. The pad hears one frame, checks its MIC and that the battery is above 2.90 V, and stores the counter it saw, $c_{init}$, with the time $t_{pair}$. At any later time $t$ the server expects

$$\hat{c}_u(t) = \left(c_{init} + \left\lfloor \frac{t - t_{pair}}{900} \right\rfloor\right) \bmod 65536, \tag{6}$$

and keeps the identifiers of slots $\hat{c}_u - 1$, $\hat{c}_u$, $\hat{c}_u + 1$ in a hash map, which covers crystal drift of $\pm 20$ ppm. Among 10,000 wearables there is about a 10% chance at any moment that two entries share a 32-bit identifier, so the map holds a list per identifier and separates a clash by the counter.

**Algorithm 1: verification and location at the server**

**Input:** pole, Rolling_ID, $c$, RSSI, F_stat, and for an SOS the whole frame.
1. Look up Rolling_ID. If there is no entry whose slot equals $c$ within $\{\hat{c}_u - 1, \hat{c}_u, \hat{c}_u + 1\}$: count it against the pole and **drop**.
2. If the SOS bit is set: recompute the MIC over bytes 2–10 with $K_u$. If it differs: count a forgery and **drop**. Otherwise raise the **priority alarm**.
3. Among reports for $u$ in the last 15 s, place $u$ at the pole with the strongest signal.
4. If a wearable appears to have moved faster than any vehicle on the track, flag the sighting as suspect.

### 4.5 What the protocol gives, and what it does not

**Unlinkability against $\mathcal{A}_1$.** AES-CMAC is a pseudorandom function [3], so without $K_u$ the identifiers and addresses of different slots are indistinguishable from random strings. Within one 15-minute slot a wearable can be followed.

**Unforgeable alarms against $\mathcal{A}_2$ and $\mathcal{A}_3$.** A forged SOS or a forged wake token passes with probability $2^{-32}$ or $2^{-24}$ per attempt. A pole forwards at most ten SOS frames a minute and a wearable accepts at most one command in 30 s.

**A stated limit: replay of ordinary sightings.** Arrival reports are 8 bytes and carry no MIC, and a pole has no key with which to check one. A frame recorded at one pole and replayed at another inside its three-slot window, up to 45 minutes, is therefore accepted as a sighting. It cannot raise an alarm, and step 4 flags it when the implied speed is impossible, but it can mislead. Carrying the MIC in arrival reports would close this at a cost of 4 bytes a wearable.

## 5 Edge filtering and capacity

### 5.1 Time on air and collisions

In the IN865 band [1, 8] the design uses $M = 8$ channels of 125 kHz at coding rate 4/5 with an 8-symbol preamble. A symbol lasts $T_{sym} = 2^{SF}/BW$, 1.024 ms at SF7. For an application payload of $B$ bytes the LoRaWAN frame is $13 + B$ bytes and its time on air is [2]

$$T(B) = T_{sym}\Big(12.25 + 8 + \max\Big(\Big\lceil \frac{8(13+B) - 4\,SF + 44}{4\,SF} \Big\rceil \cdot 5,\ 0\Big)\Big). \tag{7}$$

At SF7 this gives 66.82 ms for an SOS (15 or 16 bytes), 138.50 ms for a 64-byte heartbeat and 338.18 ms for a 200-byte chunk of 25 arrivals. Uplinks are unsynchronised, so they collide as in pure ALOHA: with offered load $G$ per channel a frame is lost with probability $1 - e^{-2G}$. Capture and the partial orthogonality of spreading factors are ignored, which makes the figures pessimistic.

### 5.2 The pole's state machine

A raw list of 4-byte identifiers every minute would exceed 222 bytes as soon as more than 55 people shelter at one pole. Each pole therefore runs Algorithm 2 over a table of 1,024 entries.

**Algorithm 2: edge filter at a pole**

**On each advertisement** with signal strength $r$:
1. Ignore it unless it is 15 bytes with Company_ID `0xFF5A`.
2. If the SOS bit is set, fewer than 10 SOS were sent this minute, and this identifier was not forwarded in the last 90 s: **send now**, confirmed, on port 10, the frame and one byte of $r$; repeat at SF9 if unacknowledged.
3. If $r$ is below the gate ($-75$ dBm on tracks, $-65$ dBm in hubs): stop.
4. Update the entry: $\mathrm{RSSI} \leftarrow 0.75\,\mathrm{RSSI} + 0.25\,r$, count the hit, note the time.
5. When the hits reach $H$ ($H = 1$ on tracks, 3 in hubs) and the arrival is not yet reported: queue the 8-byte tuple (identifier, $c$, RSSI, flags) and mark it reported.

**Every 15 s**, after a random 200–1500 ms:
6. Move entries unheard for 90 s to the departure queue.
7. Send arrivals in chunks of at most 25 tuples (200 bytes) on port 1.
8. Send departures, 4 bytes each, at most 50 a frame, on port 2.

**Every 60 s:**
9. Set three bits per reported identifier in a 512-bit filter, at $(\mathrm{CRC16} + j \cdot \mathrm{FNV1a}) \bmod 512$, $j = 0, 1, 2$, and send the 64 bytes on port 3.

The gate of $-75$ dBm corresponds, by Eq. 2, to about 22 m, which is the "core zone" a pole reports as its own. The appended signal-strength byte on port 10 is new in this revision: without it the server cannot tell which of several poles that hear an SOS is nearest.

### 5.3 Capacity of a gateway cell

Consider $N$ wearables in a cell of $K = 60$ poles, each reaching a new pole every $\tau = 55$ s. A pole then sees Poisson arrivals at $N/(\tau K)$ per second and holds $N/K$ wearables on average. Table 4 gives the offered load and loss for four designs, with every stream of each design counted. A Monte Carlo simulation that puts every frame of an hour on air, 695,304 frames in all, agrees with the loss formula to within 1.7 percentage points.

Three things follow. First, batching does most of the work: from 57.6% loss to 25.3% at 3,000 wearables. Second, the Bloom heartbeat is what keeps every frame legal: at 3,000 wearables a raw list would be too long for one frame at 22% of poles, while the filter is

**Table 4** Offered load $G$ and frame loss, in percent, as "$G$ / loss", for one 8-channel cell of 60 poles at SF7 against the number of wearables in the cell (calculated).

| Forwarding design | 500 | 1,000 | 2,000 | 3,000 | Largest frame |
|---|---|---|---|---|---|
| Raw forwarding (2 s pings) | 176.8 / 97.1 | 353.6 / 99.9 | 707.2 / 100.0 | 1060.8 / 100.0 | 7 B |
| Single-event arrivals + raw ID heartbeat | 7.6 / 14.1 | 14.6 / 25.4 | 28.7 / 43.7 | 42.9 / 57.6 | over 222 B |
| 15 s batching + raw ID heartbeat | 4.5 / 8.6 | 6.7 / 12.5 | 10.6 / 19.0 | 14.6 / 25.3 | over 222 B |
| Rakshasutra as specified | 7.7 / 14.3 | 10.2 / 18.4 | 14.2 / 24.7 | 18.2 / 30.5 | $\leq$ 200 B |
| Rakshasutra without departure lists | 5.1 / 9.7 | 6.6 / 12.4 | 9.3 / 17.0 | 12.0 / 21.3 | $\leq$ 200 B |

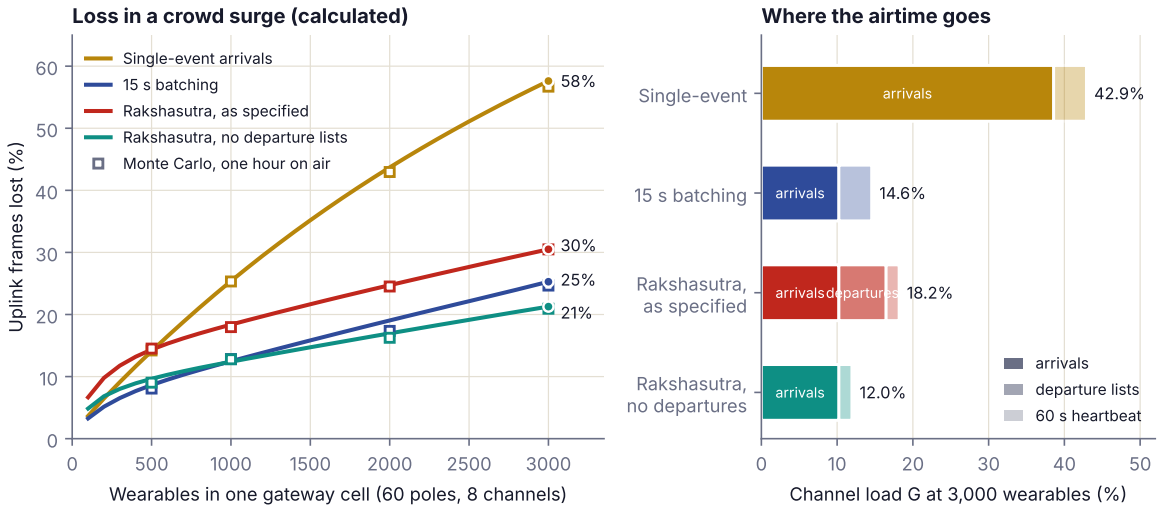



**Figure 2** Left: frame loss against the size of the surge; squares are the Monte Carlo check. Right: where the airtime goes at 3,000 wearables.

always 64 bytes; it also needs less airtime than raw lists once a cell holds more than about 960 wearables. Third, the departure lists of step 8 cost more than the heartbeat saves. As specified the design carries 25% more airtime than plain batching and loses 30.5% of frames. Without departure lists, inferring departure from the next arrival and the heartbeat, it carries 18% less and loses 21.3%. Against one uplink per arrival the airtime falls by 57.6% as specified and by 72.1% without departure lists.

In the deployed layout a gateway serves about 32 LoRa poles. With 5,000 wearables spread evenly over the route a cell holds 637 of them and loses 11.0% of frames, or 7.3% without departure lists.

A filter of 512 bits with three probes gives 1.6% false positives with 50 wearables at a pole, 8.7% with 100 and 57% with 300; it holds about 78 at a 5% error rate. The server therefore uses a heartbeat only to confirm wearables it already places at that pole. The densest poles are on fibre and can send a larger filter.

## 6 Alerts and moving people

### 6.1 Why Class A is too slow

A Class A design needs 15 s to batch an arrival, 90 s to confirm a separation, up to 60 s for a downlink window and 8 s to connect to the band: 173 s in the worst case. At 1.25 m/s a parent walks 216 m in that time, past several poles. Rakshasutra uses two layers instead.

### 6.2 Layer 1: a direct tether

At pairing the pad writes the band's next 96 identifiers, 384 bytes covering 24 hours, into the parent tag. The tag scans periodically, and if it misses the band four times in a row it sounds its own buzzer, with no network involved.

> **Two corrections to the first draft.** It specified a 5 ms scan every 5 s. The band advertises every 800 ms, so a 5 ms window contains a whole advertisement only 0.48% of the time, and four scans in a row all miss 98% of the time with the child beside the parent (simulated: in false alarm 98% of the time). A window of 811 ms cannot miss, but costs 746 $\mu$A and would empty the parent tag in about 1.9 months; synchronised scanning is the better route and is future work. Second, the signal is lost at the edge of radio range, about 174 m by Eq. 2, not at 25–30 m. To alert at 30 m the tag must treat a signal weaker than about $-78$ dBm as a miss.

### 6.3 Layer 2: Class C multicast

When the server finds $d_G \geq 200$ m between a band and its parent tag, or a guard starts a search, it sends an 8-byte wake token to the poles within two hops of the wearable, all of which listen continuously in Class C. They broadcast it as a connectionless advertisement and the wearable, which opens a short scan window every 4 s, sounds its buzzer. The token is the identifier, a command byte and 24 bits of AES-CMAC. The design target is 2.5 to 5 s from decision to buzzer; it has not been simulated or measured, and the wearable's short scan window raises the same timing question as the tether.

### 6.4 Passengers at 20 km/h

An electric vehicle at 5.56 m/s crosses a pole's 25 m core zone in 4.5 s. A band that slows its advertising to one every 2 to 5 s when its accelerometer sees no motion sends at most two or three frames in that time, so a three-hit rule reports it on at most 25% of passes. With a fixed 800 ms interval the pass holds five or six frames, and with $H = 1$ one decoded frame is enough. If a frame is lost only when another wearable in range is on air at the same moment, the capture rate is 99.7% with 150 other wearables in range of the pole and 95.0% with 300 (calculated; Fig. 3).

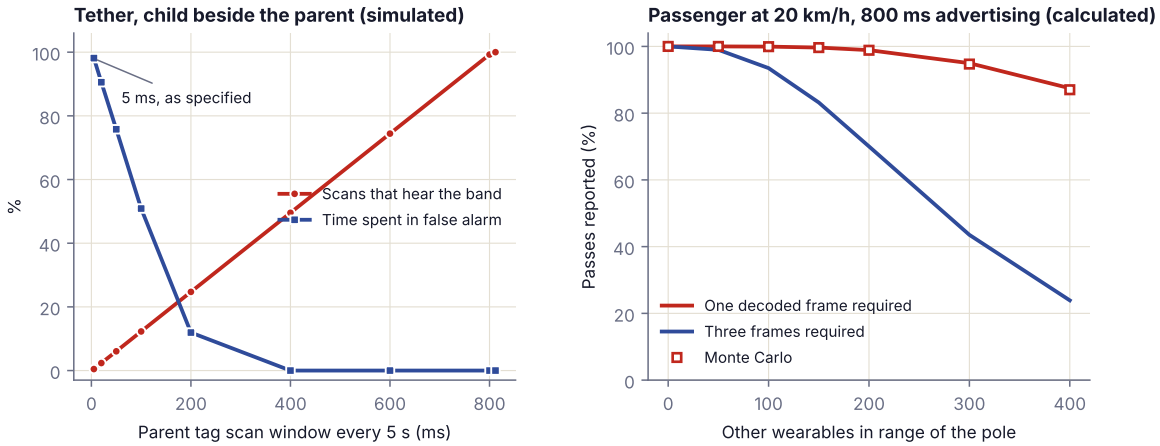



**Figure 3** Left: the tether scan window against how often the parent tag hears the band and how long it spends in false alarm (simulated). Right: capture of a passenger at 20 km/h against the crowd around the pole.

### 6.5 An SOS, end to end

Pressing the button sets the SOS bit and the 200 ms interval. A pole forwards the frame at once. Under a 3,000-wearable surge the first LoRa attempt is lost 23% of the time; a repeat follows at SF9 after an assumed 3 s. In simulation the alarm reaches the control room in a

median of 0.5 s, within 4 s in 92% of cases, and within 7.0 s in 95%. On fibre poles, where the densest crowds are, 95% arrive within 0.6 s.

### 6.6 Backend

Gateways and fibre poles publish to an MQTT broker; identity resolution and the graph distance run in memory; a worker writes batches of 500 to 2,500 records to a time-series database every 5 s. These components are specified, not built.

## 7 Power

**Table 5** Wearable current on a 620 mAh CR2450, as tabled in the first draft (design values).

| State | Interval | Peak, mA | On, ms | Mean, $\mu$A |
|---|---|---|---|---|
| Sleep, 32.768 kHz clock and RAM retention | continuous | 0.0018 | – | 1.80 |
| Signed advertisement | every 0.8 s | 14.2 | 1.25 | 22.19 |
| Scan window for a wake token | every 4 s | 4.6 | 5 | 5.75 |
| AES-CMAC slot rotation | every 900 s | 3.8 | 0.45 | 0.00 |
| Total on the trek | | | | 29.74 |

Table 5 sums to 29.74 $\mu$A. At 14 hours a day on the trek and 10 on the storage rack this gives 48.2 months at 25 °C, and 33.8 months at $-5$ °C where the cell loses 30% of its capacity. But 1.25 ms per advertisement is the length of a 1M PHY event on three channels. On Coded PHY, which the range in Section 3 assumes, the same 15 bytes take 5.95 ms: three `ADV_EXT_IND` packets and one `AUX_ADV_IND`. The mean current is then 113.2 $\mu$A and the life 12.8 months warm and 9.0 months cold (Fig. 4).

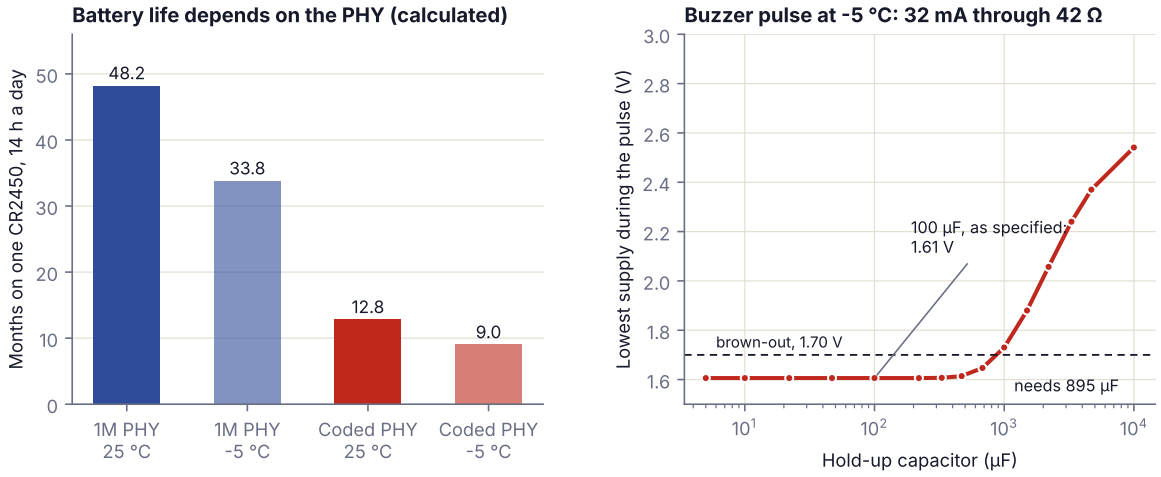



**Figure 4** Left: battery life by PHY and temperature. Right: lowest supply voltage during a buzzer pulse in the cold against the size of the hold-up capacitor (calculated).

**The cold buzzer pulse.** At $-5$ °C the cell's internal resistance rises to about 42 Ω. A 32 mA buzzer and LED pulse then drops $0.032 \times 42 = 1.34$ V, taking a 2.95 V cell to 1.61 V, below the 1.70 V minimum of the radio chip. The first draft added a 100 $\mu$F capacitor and a 20% duty cycle and reasoned from the average current. That does not hold: 100 $\mu$F behind 42 Ω has a time constant of 4.2 ms, far shorter than a 100 ms pulse, so the supply still falls to 1.61 V. Either the capacitor must be at least 895 $\mu$F, or the pulse current must stay below 29.8 mA.

**Choice of chip.** The first draft named the nRF52811 with $+8$ dBm output. That part is specified to $+4$ dBm; $+8$ dBm needs a part such as the nRF52820 or nRF52833. This is to

be confirmed against the datasheets before the bill of materials is fixed, and the currents in Table 5 re-derived for the chosen part.

## 8 Law, ethics and physical safety

**Data protection.** Section 9 of India's Digital Personal Data Protection Act, 2023 [7] requires verifiable consent from a parent or guardian before a child's data are processed, and forbids tracking for behavioural monitoring or advertising. Rakshasutra records consent at registration, keeps only a first name, a band identifier, a guardian's phone number and the card number, sends no personal data over the air, and deletes location trails within 24 to 72 hours of the band's return.

**Kiosks.** Lookup kiosks in the phone-free zones respond only to the family's own card or parent tag, show only the members bound to it, do not search by name, lock after three failed scans, log every query and blank after 30 s.

**Button cells.** A swallowed coin cell can cause severe internal burns [14]. The band is welded shut or closed with captive tool-only screws, has no battery door, and uses a non-porous strap that is disinfected between uses.

## 9 What changed since the first draft

The first draft of this paper (October 2026) called the system SurakshaNet in its text and described its results as empirical. Neither was right: the name is Rakshasutra, and nothing has been measured. For this revision every checkable number was recomputed by the open code. Of 42 claims, 16 were reproduced, 16 differ and now carry the model's value, 6 need a design fix, and 4 cannot be settled without a datasheet or a measurement. The full list is in `AUDIT.md` in the repository. The main changes:

- Loss in a 3,000-wearable surge is 21.3–30.5%, not 15.8%; the first draft counted only part of the traffic of Algorithm 2. Departure lists should be dropped.
- The airtime saving over single-event forwarding is 57.6–72.1%, not 82.4%.
- The tether cannot work with a 5 ms scan window, and alerts at the edge of radio range unless it uses a signal-strength gate.
- A 100 $\mu$F capacitor does not prevent brown-out during the cold buzzer pulse.
- Battery life on Coded PHY is 9.0–12.8 months, not "over 11 months" in the cold.
- Bluetooth ranges from Eq. 2 are longer than first tabled; the 50 m margin is 20.3 dB, not 28.6 dB.
- Ordinary sightings can be replayed inside their slot window; only alarms are fully authenticated.
- An SOS arrives within 4 s in 92% of simulated cases under a surge, not in all.

## 10 Conclusion

Rakshasutra combines a signed, rotating Bluetooth advertisement with poles that filter at the edge and forward over LoRaWAN, to follow vulnerable pilgrims between the gates of a covered mountain route without phones or cellular coverage. The calculations support the core of the design: frames stay within the payload limit at any density, batching and a Bloom heartbeat cut airtime by well over half, passengers in vehicles are captured, and forged alarms are rejected. They also show where it must change before hardware is built: the tether's scanning, the departure lists, the buzzer's power supply and the choice of radio chip. The next step is measurement: roof loss and range under a real shelter, then a pilot on one zone. Later work may extend the approach to tunnel construction sites and to temporary gatherings such as the Kumbh Mela.

## References

[1] LoRa Alliance, *RP002-1.0.4 LoRaWAN Regional Parameters*, 2022.

[2] Semtech, *LoRa Modem Designer's Guide*, AN1200.13.

[3] J. Song, R. Poovendran, J. Lee, T. Iwata, "The AES-CMAC algorithm," RFC 4493, IETF, 2006.

[4] Bluetooth SIG, *Bluetooth Core Specification*, version 5.1, 2019.

[5] B. H. Bloom, "Space/time trade-offs in hash coding with allowable errors," *Commun. ACM* 13(7):422–426, 1970.

[6] A. Kirsch, M. Mitzenmacher, "Less hashing, same performance: building a better Bloom filter," *Random Struct. Algorithms* 33(2):187–218, 2008.

[7] Government of India, *The Digital Personal Data Protection Act, 2023* (No. 22 of 2023), Section 9.

[8] Wireless Planning and Coordination Wing, Government of India, rules for low-power equipment in the 865–867 MHz band.

[9] A. Augustin, J. Yi, T. Clausen, W. M. Townsley, "A study of LoRa: long range and low power networks for the Internet of Things," *Sensors* 16(9):1466, 2016.

[10] M. Centenaro, L. Vangelista, A. Zanella, M. Zorzi, "Long-range communications in unlicensed bands," *IEEE Wireless Commun.* 23(5):60–67, 2016.

[11] F. Adelantado et al., "Understanding the limits of LoRaWAN," *IEEE Commun. Mag.* 55(9):34–40, 2017.

[12] A. Heinrich, M. Stute, T. Kornhuber, M. Hollick, "Who can find my devices? Security and privacy of Apple's crowd-sourced Bluetooth location tracking system," *Proc. Privacy Enhancing Technol.* 2021(3):227–245.

[13] Nordic Semiconductor, *nRF52811 Product Specification*.

[14] IEC 60086-4, *Primary batteries, Part 4: Safety of lithium batteries*.